A Needle in a Data Haystack

League of Legends

Data Science Report

Finding the Needle in Ranked Play

Behavioral and Strategic Patterns in League of Legends

GitHub | Streamlit (Demo)

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Finding the needle in ranked play

Domain & Motivation

League of Legends (LoL) is a competitive multiplayer game in which two teams of five players compete in repeated matches. Ranked play produces rich longitudinal data: the same players appear across many games, allowing analysis not only of what happens inside a match, but also of behavioral patterns between matches.

Our project uses this temporal structure to study patterns that are difficult to detect from individual games alone. In particular, we distinguish between variables known before a target match and statistics produced during it, preventing target-match information leakage in predictive analyses.

Data

Our primary dataset is a collection of Riot Match-V5-style records covering **North America (NA), Korea (KR), and Europe (EU)**. The raw data occupy approximately **75 GB uncompressed**. After streaming extraction and deduplication, the processed corpus contains **497,102 unique matches, 4,971,020 participant rows, and 994,204 team rows**.

We converted the raw JSON data to compressed Parquet tables with separate match-, participant-, team-, and champion-ban-level records, allowing efficient analysis of the full corpus. The source dataset is: https://www.kaggle.com/datasets/omaracornejo/matches-raw-euw

A major sampling issue is that matches were collected by crawling the histories of predefined seed players. Thus, many players appearing in a downloaded match are incidental teammates or opponents and should not automatically be treated as longitudinally sampled players. We reconstructed an authoritative tracked-player cohort using the original seed lists together with stable Riot PUUID evidence from the accompanying database. The final cohort contains 11,169 player-region identities and covers 100% of NA matches, 100% of KR matches, and 99.95% of EU matches.

Before analysis, we verified sampled raw PUUID-to-processed-ID mappings, participant counts, duplicate IDs, timestamps, missingness, and chronological ordering. The final longitudinal data contain no duplicate (player, match) observations and no negative inter-match gaps. Remaining limitations are that the data are crawler-based rather than a random sample of all LoL players, histories are limited to the observed crawl window, and very short/remake-like games can distort performance measures. These issues are addressed through explicit filtering and robustness analysis rather than silent removal.

Temporal Behavior & Next-Match Performance

Problem & Hypotheses

We investigate whether **session depth, recent ranked-game volume, and post-loss requeue timing** are associated with performance in a player's subsequent Ranked Solo/Duo match. We test whether performance worsens with deeper sessions (H1), rapid requeueing after a loss (H2), and greater recent volume (H3), and whether these behavioral variables add predictive information beyond historical player performance (H4). Because the data contain no psychological measurements, these variables are treated as observable behavioral patterns rather than direct measures of fatigue or “tilt.”

Solution / Method

For each authoritative tracked player, matches were ordered chronologically and converted to target-centric observations containing only information available before the target match. The primary sample comprised **1,146,681 Ranked Solo/Duo targets lasting ≥10 minutes** across NA, KR, and EU. Inter-match gaps showed a dense short-gap component followed by a long tail (Fig. 1);

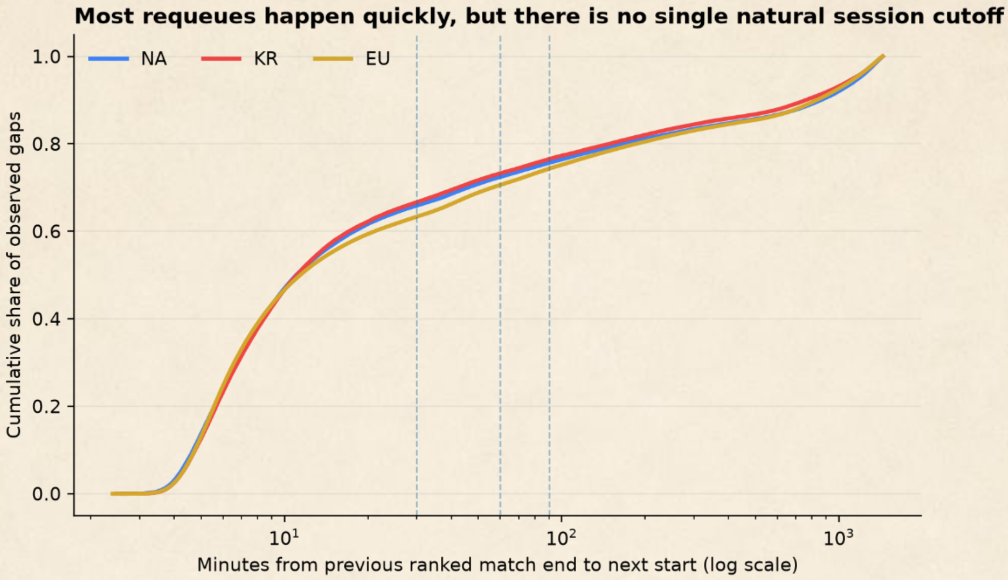

Figure 1. Distribution of ranked inter-match gaps across regions. Most observed requeues occur after relatively short gaps, followed by a long tail. We therefore use a conservative 30-minute primary session boundary and test 45-, 60-, and 90-minute alternatives.

Recent volume used a **6-hour** primary window with 3/12/24-hour alternatives. We estimated within-player linear probability models with player fixed effects, pre-target history/patch controls, and standard errors clustered by player and physical match. For prediction, we used entropy-based decision trees and selected max_depth and min_samples_leaf on validation data to control overfitting.

Evaluation & Setup

For H1–H3, success required a consistent, practically meaningful, cross-region association, not merely $p<.05$. We therefore examined effect sizes, 95% CIs, monotonicity, Holm-corrected tests, and sensitivity to alternative thresholds. For H4, Riot's observed target-match win/loss was the ground truth. Data were split chronologically within region into 70% train / 15% validation / 15% test, and we compared majority and historical-win-rate baselines with history-only, behavior-only, and combined trees. Evaluation used accuracy, precision, recall, F1, confusion matrices and ROC-AUC.

Results & Visualization

Raw averages did not show the expected simple fatigue pattern, motivating adjusted within-player analysis:

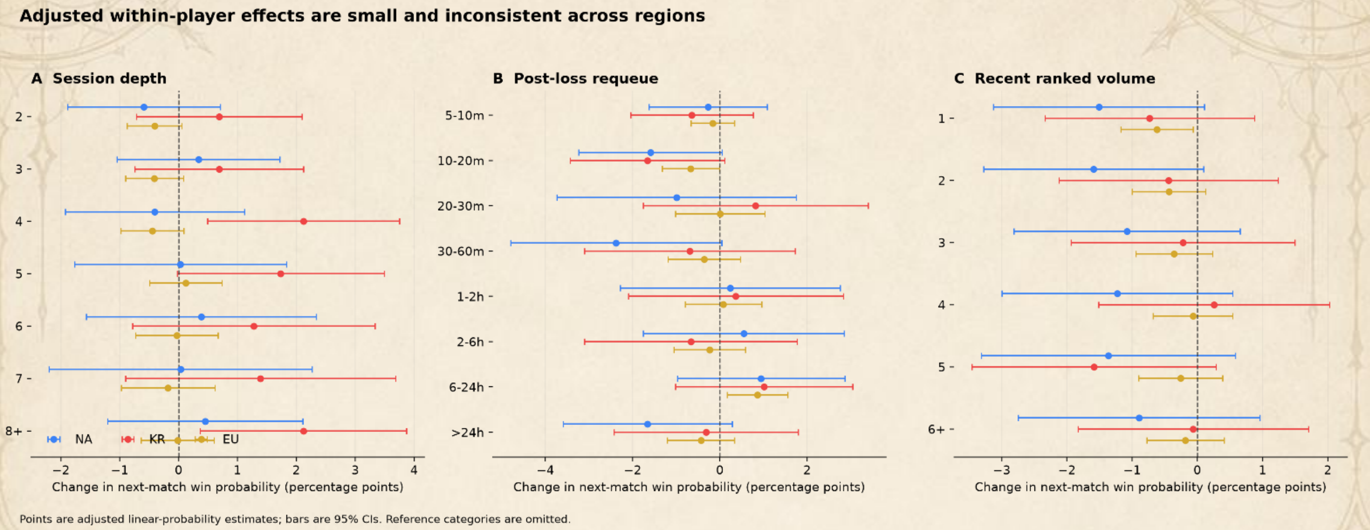

Figure 2. Adjusted within-player associations with subsequent Solo/Duo victory. Points show estimated percentage-point changes in win probability and intervals show 95% CIs. Across session depth, post-loss requeue timing, and recent ranked volume, estimates are small, non-monotonic, and inconsistent across regions.

Figure 2 summarizes the primary estimates: session depth, post-loss requeue timing and six-hour volume produced small, non-monotonic, region-dependent effects. In the continuous post-loss model, doubling the requeue gap changed next-match win probability by approximately −0.04 pp (NA), +0.11 pp (KR), and +0.03 pp (EU). No primary behavioral family retained a significant term after Holm correction.

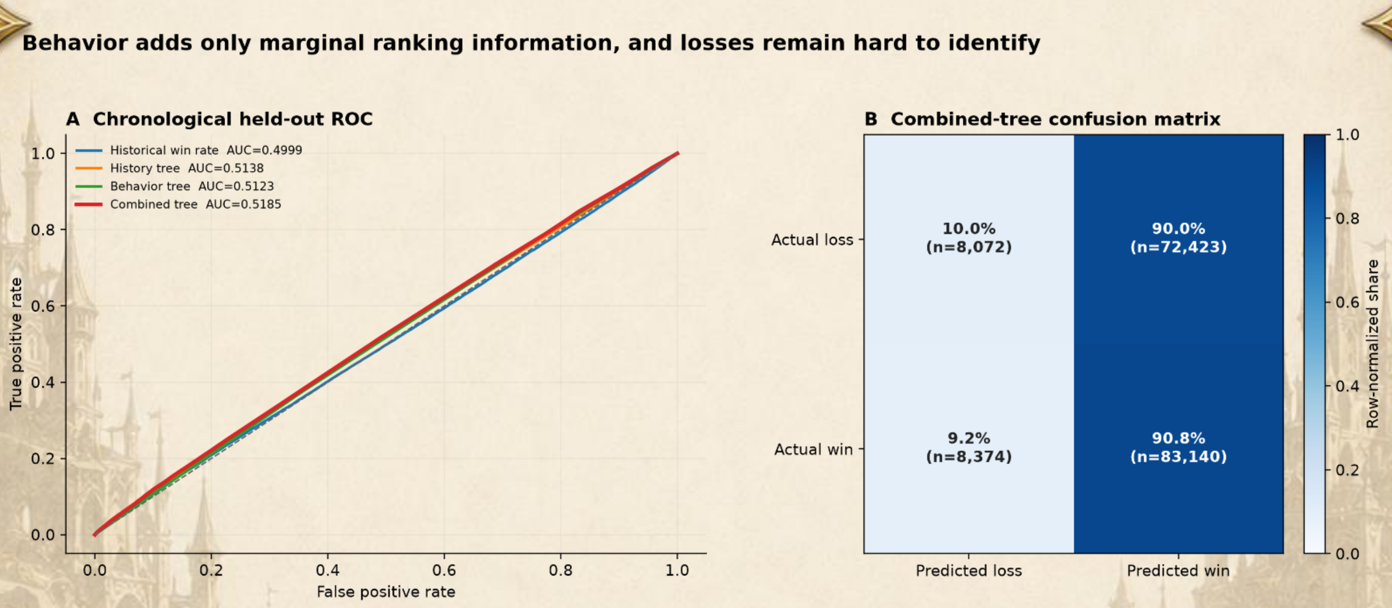

Figure 3. Chronological held-out predictive performance. The combined history + behavior tree improves ROC-AUC only slightly over the history-only model, while the normalized confusion matrix shows that the model identifies wins much more readily than losses.

Prediction was similarly weak (Fig. 3): test AUC was 0.5138 for the history tree, 0.5123 for behavior alone, and 0.5185 for the combined tree, an incremental gain of only 0.0047 over history. Accuracy remained near the majority baseline, while the confusion matrix showed poor loss discrimination.

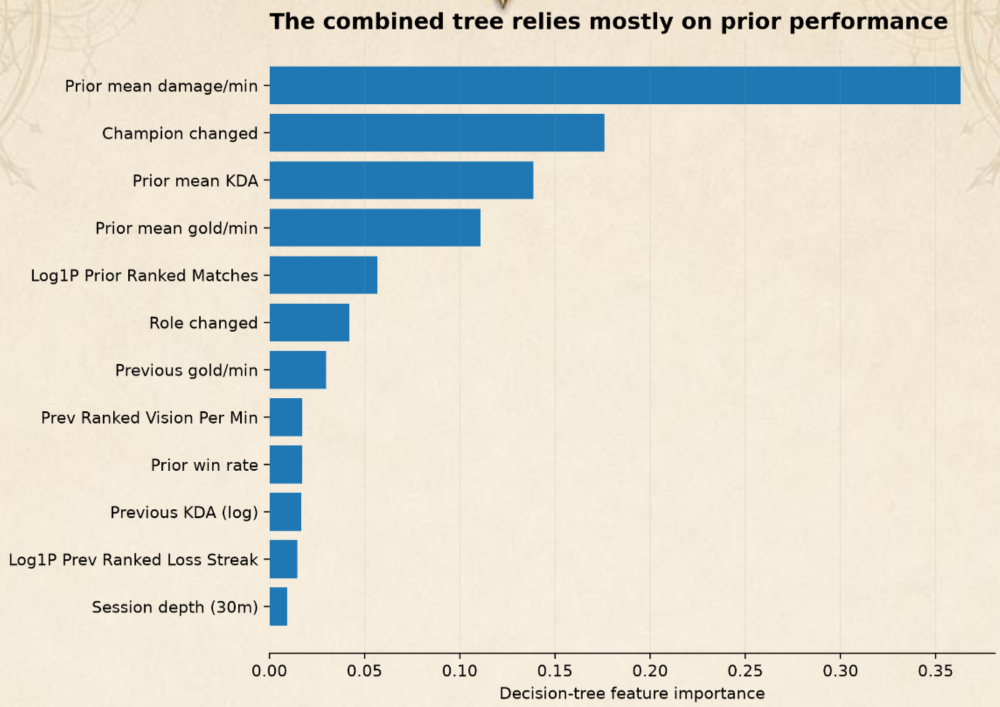

Figure 4. Feature importance of the combined entropy-based decision tree. Historical performance features account for most of the model's splitting decisions, while session and requeue-related variables contribute comparatively little.

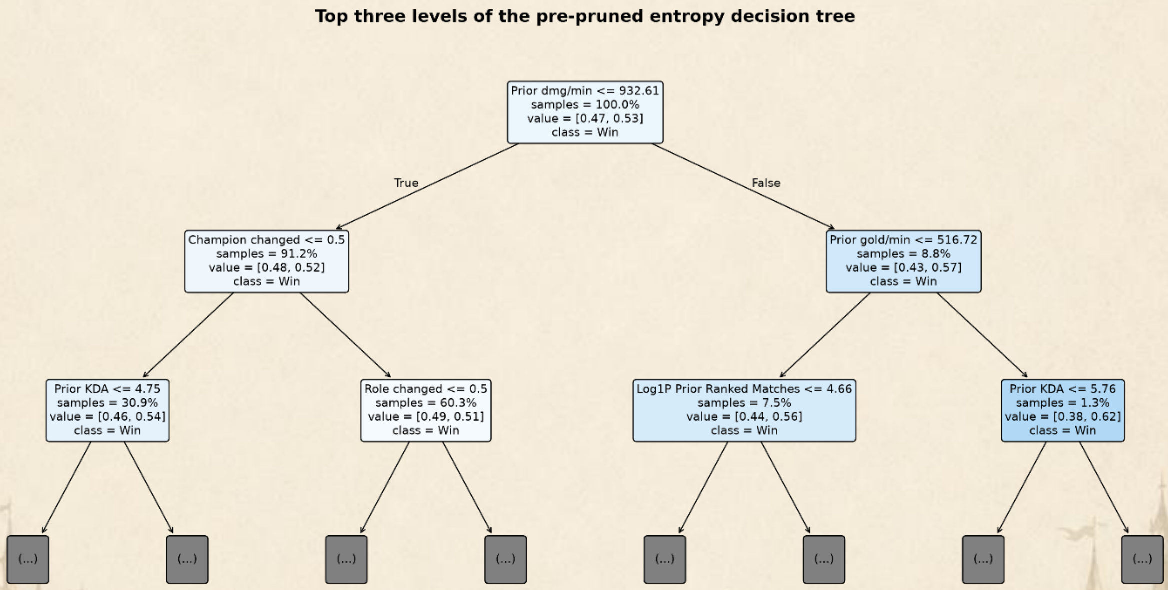

Figure 5. Top three levels of the combined entropy-based decision tree. The first splits are driven mainly by prior performance statistics, illustrating how the model separates observations before incorporating weaker behavioral features.

The entropy tree relied mainly on historical damage, KDA and gold variables; champion/role switching contributed more than session depth or raw gap (Fig. 4).

Impediments & Robustness

The main challenges were crawler-biased player sampling, arbitrary session boundaries, short/remake-like games, repeated player observations, and shared physical matches. We addressed these through PUUID-based tracked-player reconstruction, alternative session/activity thresholds, two-way clustered uncertainty, and robustness samples. Lowering the duration rule from 10 to 5 minutes changed primary effects by at most 0.09 pp, while the stricter alias-confirmed cohort produced no new stable cross-region pattern or Holm-significant primary family.

Conclusion

Overall, we find **little evidence for a universal short-term fatigue or post-loss requeue penalty**. Session depth, recent ranked volume, and rapid requeueing show small, parameter-sensitive associations rather than a stable cross-region decline in performance. Behavioral timing adds only limited predictive information, and because the data are observational, the results should not be interpreted as causal evidence about fatigue or tilt.

Champion Pairings & Combo Performance

Problem & Hypotheses

We study which champions tend to appear together on the same team and whether frequently paired champions also show stronger performance. A major difficulty is that a pair may appear often simply because both champions are individually popular. We therefore distinguish between **raw co-pick frequency** and **normalized association**, which measures whether two champions are selected together more often than expected from their individual pick rates.

We test three hypotheses: **H1:** raw frequency and normalized association identify different pairings; **H2:** some pairs are selected together substantially more often than expected; and **H3:** strong co-selection does not necessarily imply stronger pair performance.

Solution / Method

The analysis uses all valid Ranked Solo/Duo teams from matches lasting at least 10 minutes. Each five-champion team contributes its ten unordered champion pairs.

For every pair, we calculate the number of games played together and **$\log_2$(lift)**. Lift compares the observed co-pick frequency with the frequency expected if the two champions were selected independently according to their overall popularity. Values above zero therefore indicate **above-expected co-selection**.

To examine performance, we calculate each pair's win rate and a descriptive **win surplus**: the pair's observed win rate minus the average individual win rate of its two champions. We require substantial pair support before interpreting normalized association or performance, reducing the influence of rare combinations.

Evaluation & Setup

The main comparison is between raw and popularity-normalized pair rankings. If the rankings differ, raw frequency alone is insufficient for identifying unusually strong champion relationships.

Performance is evaluated separately from association using the combo landscape, which plots normalized co-selection against win surplus while also representing the number of games supporting each pair. Because this is observational data, win surplus is treated as a descriptive association rather than evidence of causal champion synergy.

Results & Visualization

Raw frequency and normalized association produce noticeably different rankings. Some combinations appear extremely often because their champions are popular, while other pairs become much more prominent after correcting for individual pick frequency. This supports H1 and H2: **popular combinations and unusually strong co-selection relationships are not equivalent**.

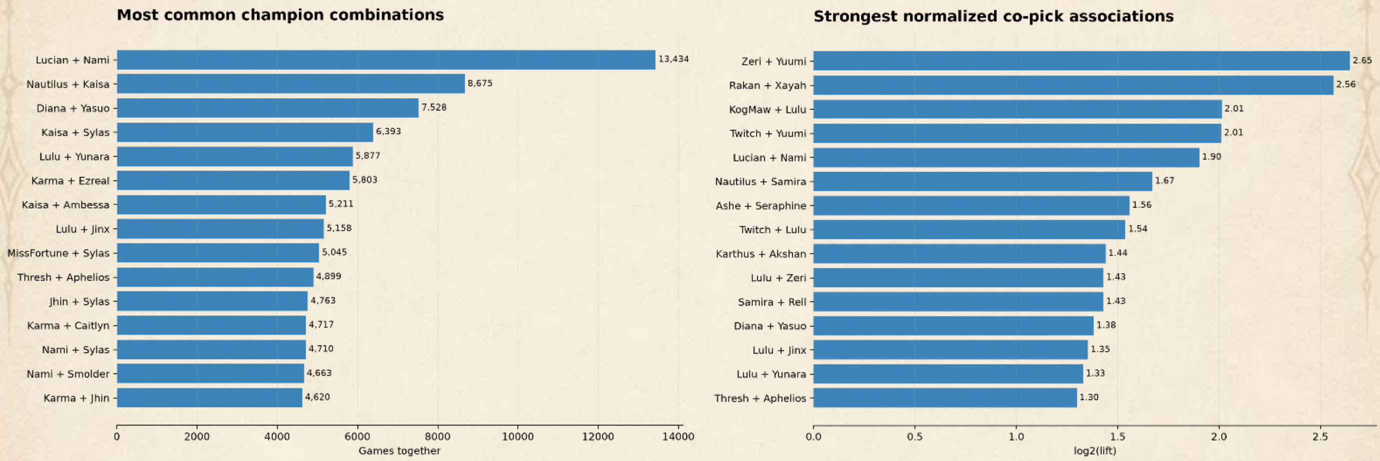

Figure 6. Raw and normalized champion co-selection. Correcting for champion popularity changes which pairings appear strongest.

The relationship between co-selection and performance is also not straightforward. In the combo landscape, highly associated pairs appear both above and below zero win surplus, while some less strongly associated pairs perform relatively well. Thus, H3 is supported: **being selected together unusually often does not automatically imply stronger outcomes**.

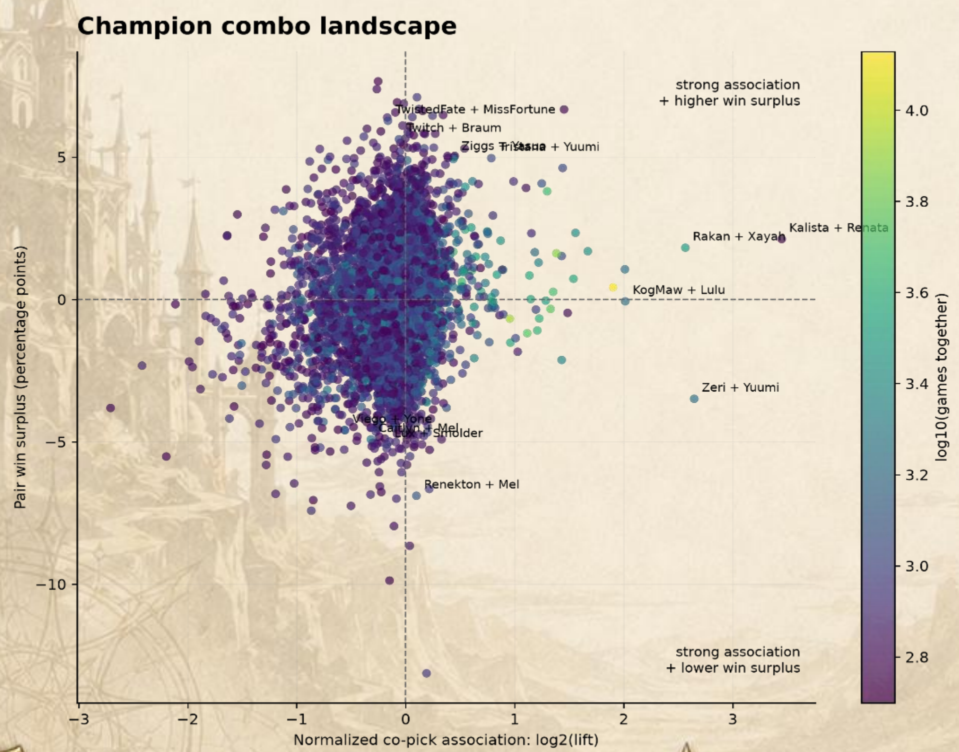

Figure 7. Champion co-selection strength versus pair win surplus. Strong association and strong performance are distinct properties.

Impediments & Robustness

The main challenges are popularity bias, rare-pair instability and the temptation to interpret every strong pairing as strategic synergy. We address these using normalized lift, minimum game-count thresholds and separate treatment of co-selection and performance.

Pair outcomes may still reflect champion strength, player skill, roles, patch effects and other confounders. Therefore, performance results are descriptive and not causal.

Conclusion

Champion pairings contain meaningful structure beyond raw popularity. Normalized association reveals combinations that are selected together disproportionately often, while the performance analysis shows that these relationships do not consistently translate into higher win surplus. **Champion popularity, co-selection preference and pair performance should therefore be treated as separate dimensions.**

Champion Network Structure & Team-Composition Communities

Problem & Hypotheses

Pairwise relationships can also be viewed as a larger **champion network**, allowing us to ask how the entire team-composition ecosystem is organized.

We investigate whether the normalized co-selection network contains meaningful communities, which champions occupy central structural positions, and whether strong pair relationships extend into recurring multi-champion groups.

We test: **H1:** the network contains interpretable community structure; **H2:** these communities are related to champion roles; **H3:** some champions are structurally central across many composition relationships; and **H4:** strong pairwise relationships extend into tightly connected multi-champion structures.

Solution / Method

Champions are represented as **nodes**, and supported above-expected co-picks form weighted **edges**, using the normalized association from Problem 2.

We apply **Louvain community detection** to identify densely connected champion groups and use modularity to describe the strength of the partition. We then compare community membership with observed Top, Jungle, Mid, ADC and Support roles.

Weighted **PageRank** identifies champions that are central not simply because they are popular, but because they connect to other important champions in the normalized network.

We also search for **maximal cliques**, where every champion in a group is strongly connected to every other member, and use clique percolation to explore overlapping structures.

Finally, we compare the graph representation with conventional unsupervised methods: **K-Means++, hierarchical clustering and DBSCAN** on champions' complete co-pick profiles.

Evaluation & Setup

Because no ground-truth community labels exist, the analysis is evaluated through **structural quality and interpretability** rather than prediction accuracy.

Louvain communities are examined using modularity and their relationship with champion roles. PageRank is interpreted as network centrality. Clique analysis tests whether strong relationships extend beyond isolated pairs.

For K-Means++, candidate values of k=2–6 are compared using within-cluster SSE and silhouette score. Hierarchical clustering and DBSCAN provide alternative views of the same co-pick profiles.

Results & Visualization

The normalized network reveals clear groups of champions rather than one homogeneous set of relationships.

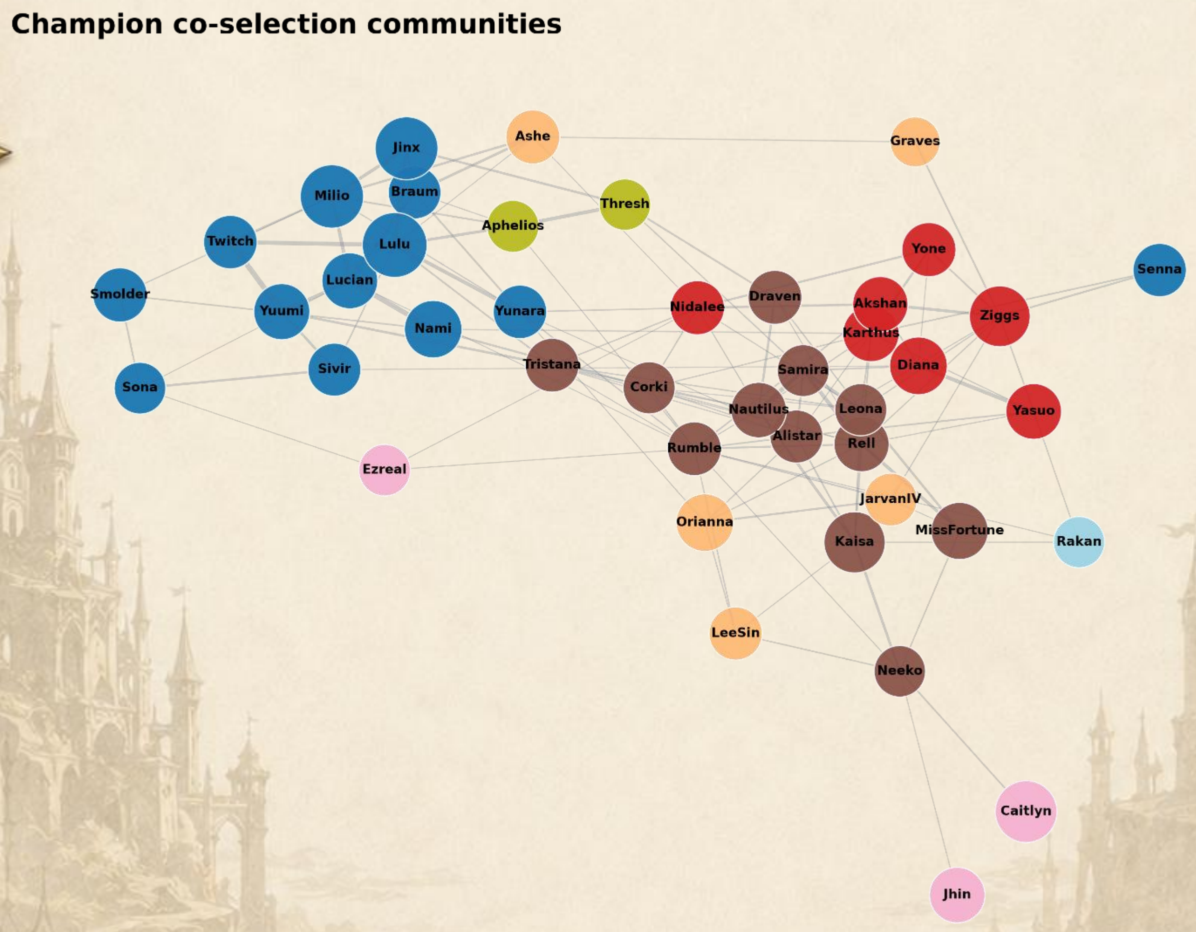

Figure 8. Louvain communities in the normalized champion co-selection network.

The communities become especially interpretable when compared with champion roles. Several groups are heavily concentrated in **ADC and Support**, while others contain much larger shares of **Top, Jungle and Mid** champions. This strongly supports H1 and H2: the detected communities are related to the functional structure of team composition rather than being arbitrary mathematical partitions.

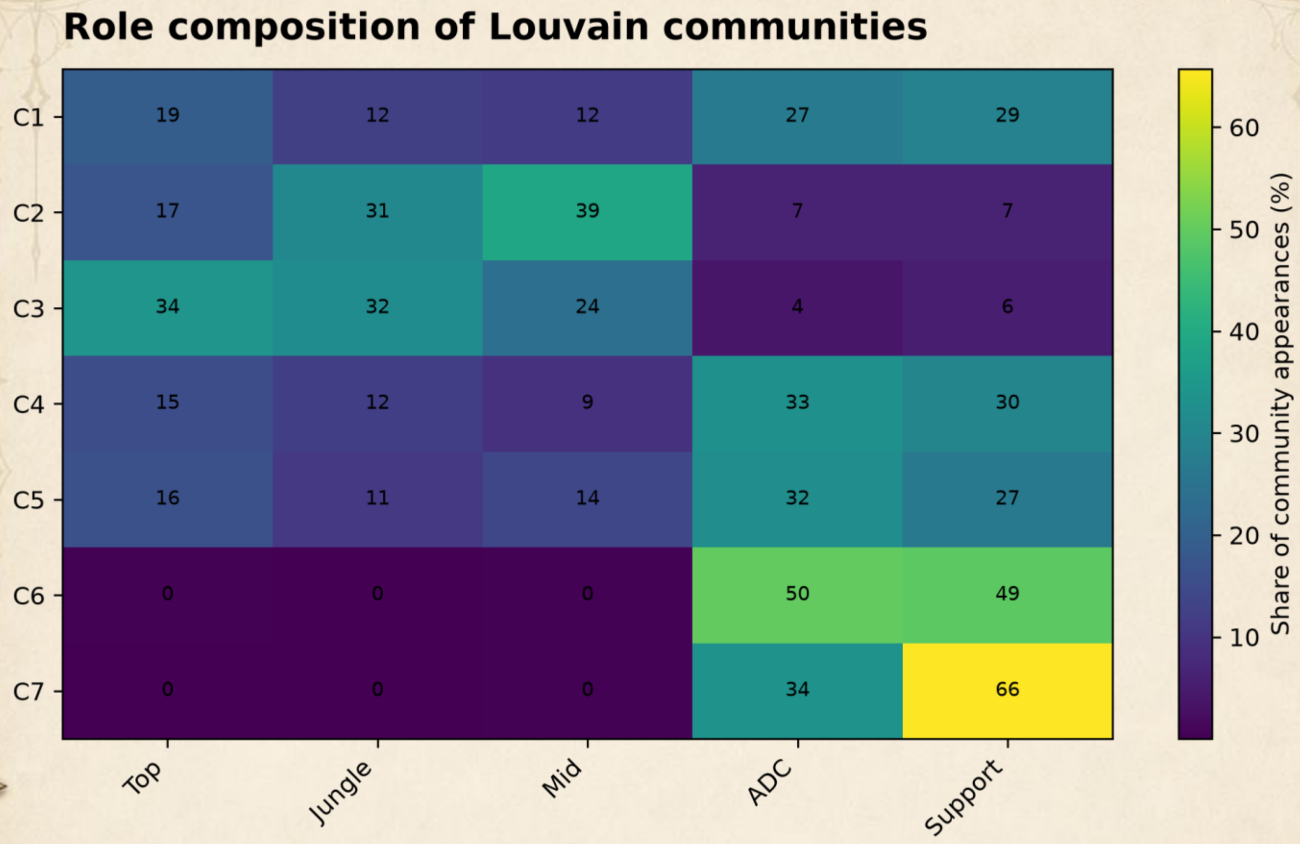

Figure 9. Role composition of Louvain communities. Communities show distinct positional profiles.

This also provides an important interpretation: some strong champion relationships may arise because their roles are naturally complementary, not necessarily because of direct gameplay synergy.

PageRank reveals another layer of structure by identifying champions connected to many important above-expected relationships. These champions occupy central positions across the composition network, supporting H3.

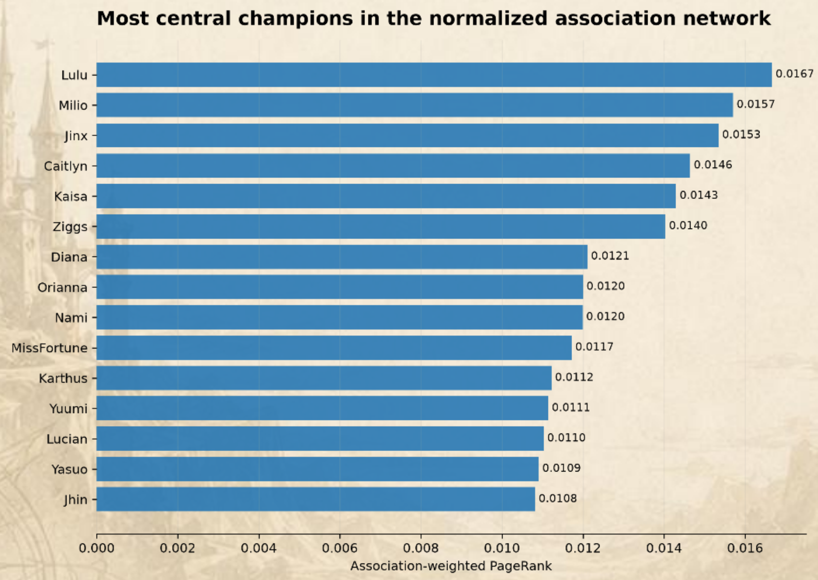

Figure 10. Association-weighted PageRank of the most structurally central champions.

Strong pairwise relationships also extend into larger groups. Maximal-clique analysis identifies champion trios in which every member is strongly associated with every other member, providing recurring **higher-order composition patterns** and supporting H4.

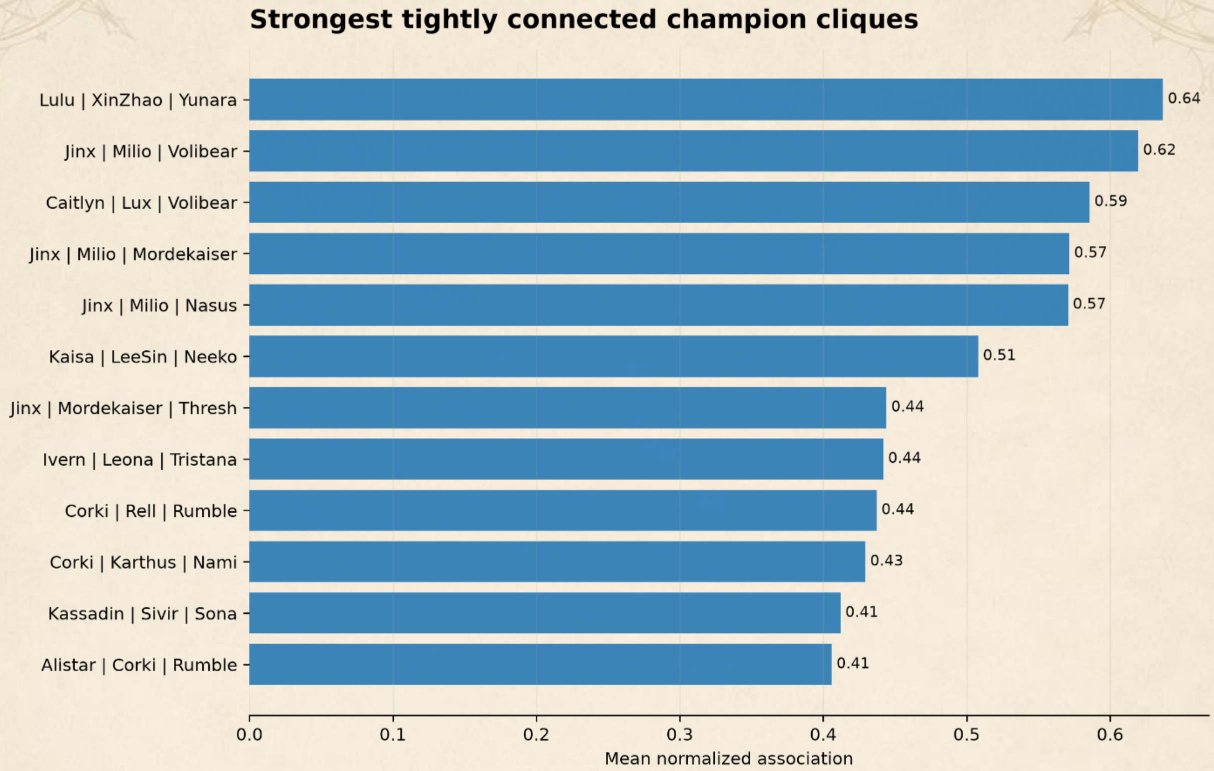

Figure 11. Strongest multi-champion cliques in the normalized co-selection network.

Impediments & Robustness

Network results depend on how edges are defined and filtered, so weak and poorly supported pair relationships are excluded before community analysis. We also compare Louvain with Girvan–Newman, clique-based methods and several conventional clustering algorithms rather than relying on one partitioning method.

The alternative methods generally produce less detailed or less interpretable structures, which supports using Louvain as the primary community representation. Role structure is also an important confounder: communities should be interpreted as **team-composition patterns**, not automatically as strategic synergy.

K-Means++ validation favored a coarse two-cluster solution, while hierarchical clustering and DBSCAN produced similarly broad or outlier-driven partitions.

Conclusion

Champion selection forms a structured network with interpretable communities, central champions and recurring multi-champion groups. Louvain communities strongly reflect team roles, while PageRank identifies champions occupying influential positions across many above-expected co-selection relationships.

Conventional clustering captures only a coarse version of this structure, suggesting that **graph-based methods are better suited to the relational nature of champion team composition**.

Future Work

Several extensions could make the project richer. For the temporal analysis, longer player histories could support questions such as churn, return behavior, rank progression, and whether behavioral effects differ by skill level, role, or champion familiarity.

The champion-composition analysis could also be made dynamic by tracking how pair associations, PageRank and communities change across patches, regions and shifts in the game meta. A more ambitious extension would model complete five-champion drafts together with roles, bans and the opposing composition, rather than studying pairs separately.

Finally, these analyses could be turned into an interactive draft and exploration tool: a user could select a champion and inspect common partners, unusually strong associations, network communities and historical performance. With richer experimental or quasi-experimental data, future work could also better distinguish genuine champion synergy from selection effects and other confounders.

Conclusion

Using nearly half a million League of Legends matches, we examined the data from three complementary perspectives. First, we found **little evidence for a universal short-term fatigue or rapid post-loss requeue penalty**, and behavioral timing added only limited predictive information beyond prior performance. Second, champion-pair analysis showed that **raw popularity, above-expected co-selection and pair performance are distinct properties**. Third, graph analysis revealed **interpretable champion communities, structurally central champions and recurring multi-champion patterns**, with graph-based methods capturing the relational structure more naturally than conventional clustering.

Overall, the project shows that seemingly simple patterns in ranked play often become more nuanced once sampling, normalization, temporal leakage and network structure are considered. The results are observational rather than causal, but together they demonstrate how multiple data-science methods can uncover complementary behavioral and strategic structure within the same large-scale game dataset.